Sampling and Intended Use

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Sampling and Intended Use before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Label Quality

Without opening the old lesson, explain one key idea from Label Quality. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Python and Table Readiness

Without opening the old lesson, explain one key idea from Python and Table Readiness. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

The ML Lifecycle

Without opening the old lesson, explain one key idea from The ML Lifecycle. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Sampling and Intended Use trace and record the first step you can explain without help.

Use W2: Interpret the audit counts for Dataset inspection and cards. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Sampling and Intended Use explanation against the supplied facts before discussing it.

Use the worked case for Dataset inspection and cards. Proposed explanation: “A complete card certifies data quality”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Changed independent case

Training plant photos are all indoors on one phone. **Intended use** includes outdoor photos on other cameras. Plan a sampling check before making a general accuracy claim.

> **My independent explanation, trace and limits**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Sampling and Intended Use target to a trace and a justified conclusion.

Training uses indoor photos from one phone; deployment uses outdoor photos from several phones. Specify two evaluation slices and explain why adding more indoor photos may not resolve the gap.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Sampling and Intended Use case and show the deciding evidence.

Training photos use indoor white backgrounds; **intended use** is outdoors.

Identify the coverage gap.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

Training photographs all show mature plants in bright light, but **intended use**rs photograph seedlings indoors. What claim is unsupported and what data is needed?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 7 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Sampling and Intended Use adds to Responsible Plant Classifier and keep the evidence.

Freeze the Plant Classifier intended-use statement and a coverage table for species, lighting, camera, and health class.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Training uses indoor photos from one phone; deployment uses outdoor photos from several phones. Specify two evaluation slices and explain why adding more indoor photos may not resolve the gap.

> **My independently reasoned answer**

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Define the row and source

> Define the row and source. Apply the supplied Sampling and Intended Use facts and keep your reasoning.

For Robot practice v1, state what one row is intended to represent, the source, the unit and the meaning of None.

> **My reasoning and evidence**

WORKBOOK · 13 OF 21

W2 Interpret the audit counts

> Interpret the audit counts. Apply the supplied Sampling and Intended Use facts and keep your reasoning.

Explain each number in 5 4 2 1. How many extra ID occurrences are there, and how many rows belong to a repeated-ID group?

My reasoning and evidence

W3 Reject an unsupported correction

> Reject an unsupported correction. Apply the supplied Sampling and Intended Use facts and keep your reasoning.

A learner changes None to 0 and -2 to 2, then reports a clean dataset. Explain why this is not a justified repair and what record should be kept.

My reasoning and evidence

W4 Inspect and document independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Inspect and document independently. Apply the supplied Sampling and Intended Use facts and keep your reasoning.

Rules supplied for this independent case

Schema for this exercise: one intended row per distinct trip ID; each row is an (ID, duration) pair. Duration is numeric minutes or None for unmeasured. Negative durations are invalid; a recorded zero is allowed. Count extra ID occurrences beyond the first and do not silently alter the raw data.

A new fictional indoor trip export has (P,4), (Q,None), (Q,None), (R,-1), (S,6), (T,0), with durations in minutes and the same **schema**.

Count raw rows, distinct IDs, missing durations, negative durations and extra ID occurrences. Identify overlap. Draft a short **dataset card** stating source, row meaning, unit, findings, unchanged version and one excluded use.

My reasoning and evidence

W5 State an evidence gap

> State an evidence gap. Apply the supplied Sampling and Intended Use facts and keep your reasoning.

Someone duplicates each row ten times and claims the export now covers more kinds of trips. Explain what changed and what did not.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Sampling and Intended Use

> Sample for the conditions in which the plant classifier will actually be used. Explain the labelled step before checking the changed condition.

Sample for the conditions in which the plant classifier will actually be used.

Training set: 12 windowsill plants photographed at midday. **Intended use** includes evening desk photos; no evening examples have been collected.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.



Follow the labelled picture with the matching explanation. Open the vector diagram

Which collection would address the intended-use gap most directly? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Responsible Plant Classifier record.

Project: Responsible Plant Classifier

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Sample for the conditions in which the plant classifier will actually be used.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Collect consented fictional or teacher-created evening examples; keep plant groups separate across development and final evaluation.

1. Identify gap

2. Declare coverage

3. Plan comparison

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Plan comparison Collect consented fictional or teacher-created evening examples; keep plant groups separate across development and final evaluation.

Read the picture in order. Collect consented fictional or teacher-created evening examples; keep plant groups separate across development and final evaluation.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Collect 12 more midday windowsill photos of the same plants. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. The evening-coverage gap remains.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The evening-coverage gap remains. More rows in the same condition do not supply **evidence** about the missing condition.

Repair a missing building block

Review Label Quality (ai-4/6). Goal: Write an observable label rule with boundaries and a review option. Short practice: Why should two annotators not vote away Z's missing **evidence**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Python and Table Readiness (ai-4/4). Goal: Specify a table **schema**. Short practice: Why should the **missing value** not silently become zero?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review The ML Lifecycle (ai-4/1). Goal: Distinguish **fitting** a rule from applying a fixed rule. Short practice: Should the team silently retune on the changed test?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

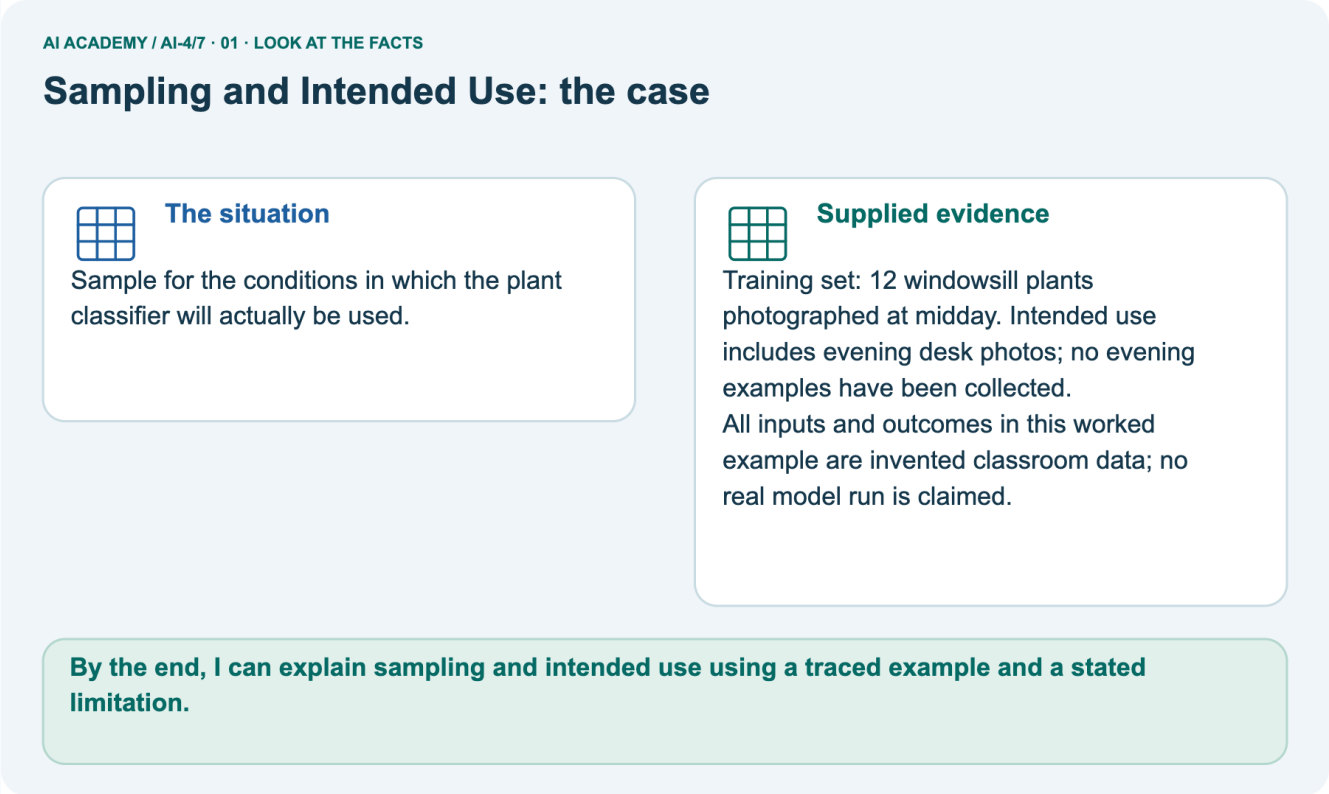
What I repaired, the evidence I kept and the next test

WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Sampling and Intended Use picture from its labels, then record a relevant limit.

01. Sampling and Intended Use: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

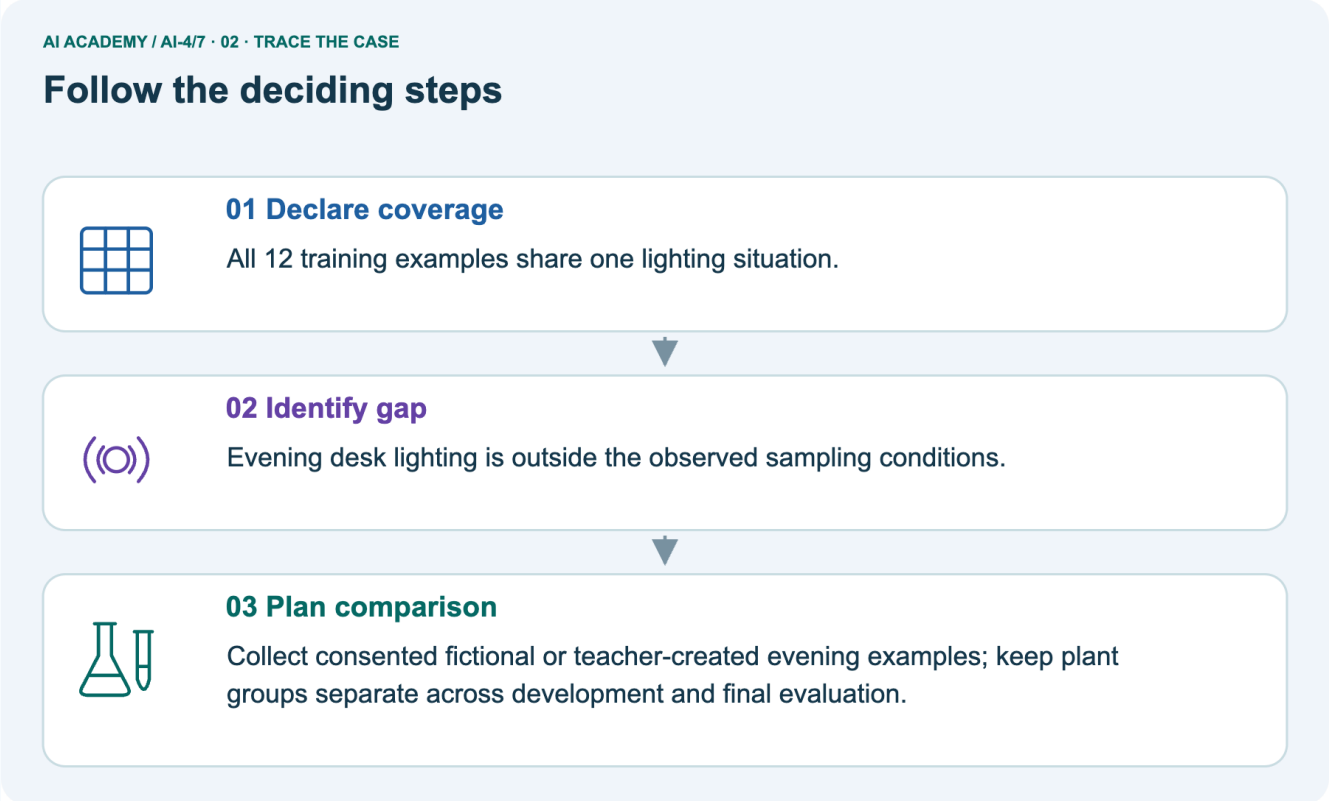
Sample for the conditions in which the plant classifier will actually be used. Supplied facts: **Training** set: 12 windowsill plants photographed at midday.

Intended use includes evening desk photos; no evening examples have been collected. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



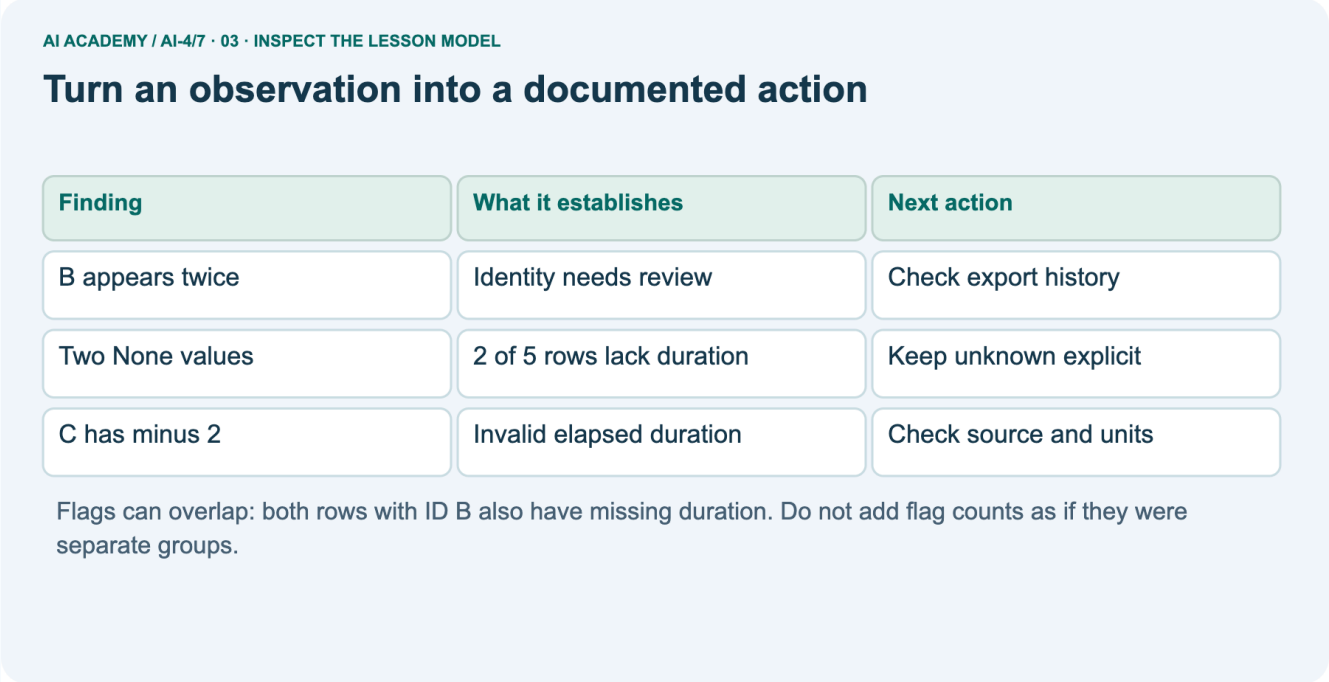
Follow the labelled picture with the matching explanation. Open the vector diagram

Declare coverage: All 12 **training** examples share one lighting situation. → Identify gap: Evening desk lighting is outside the observed sampling conditions. → Plan comparison: Collect consented fictional or teacher-created evening examples; keep plant groups separate across development and final evaluation.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Turn an observation into a documented action



Follow the labelled picture with the matching explanation. Open the vector diagram

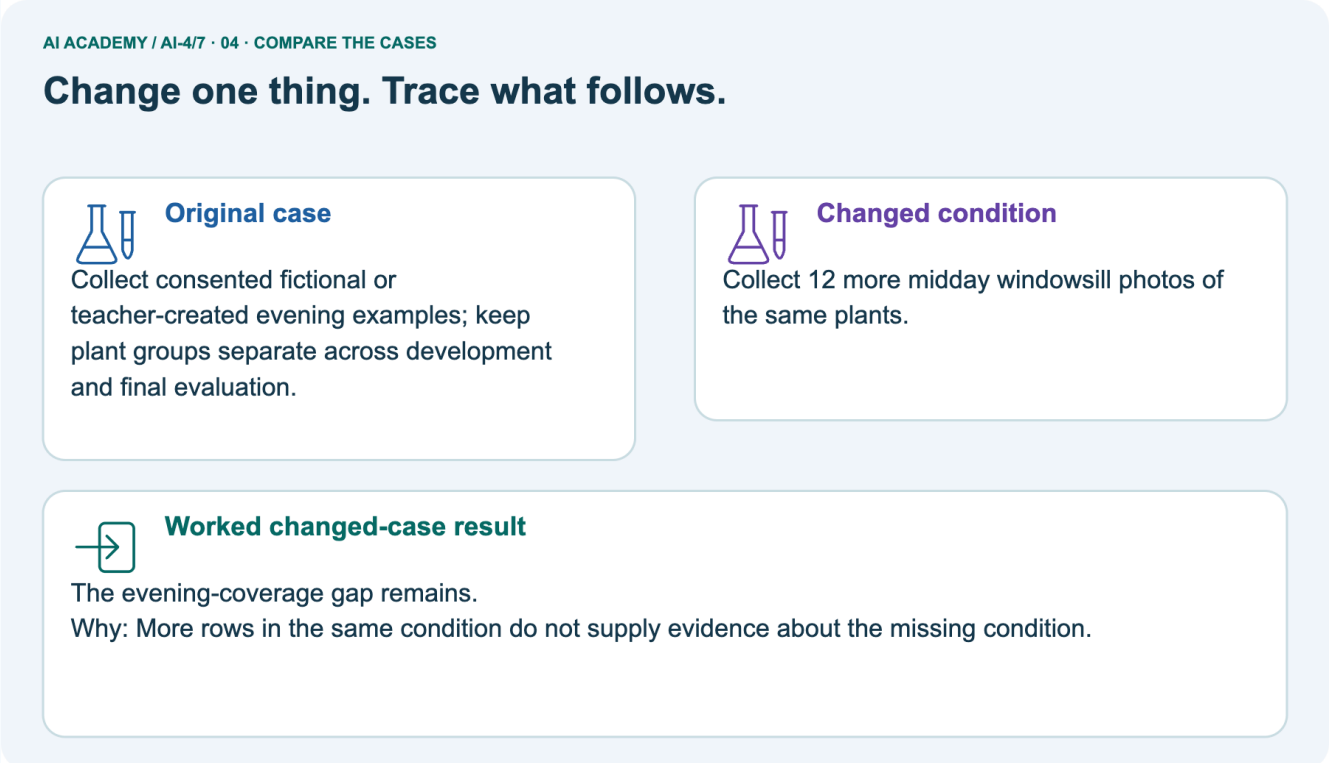
Turn an **observation** into a documented action. B appears twice | Identity needs review | Check export history. Two None values | 2 of 5 rows lack duration | Keep unknown explicit.

C has minus 2 | Invalid elapsed duration | Check source and units

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

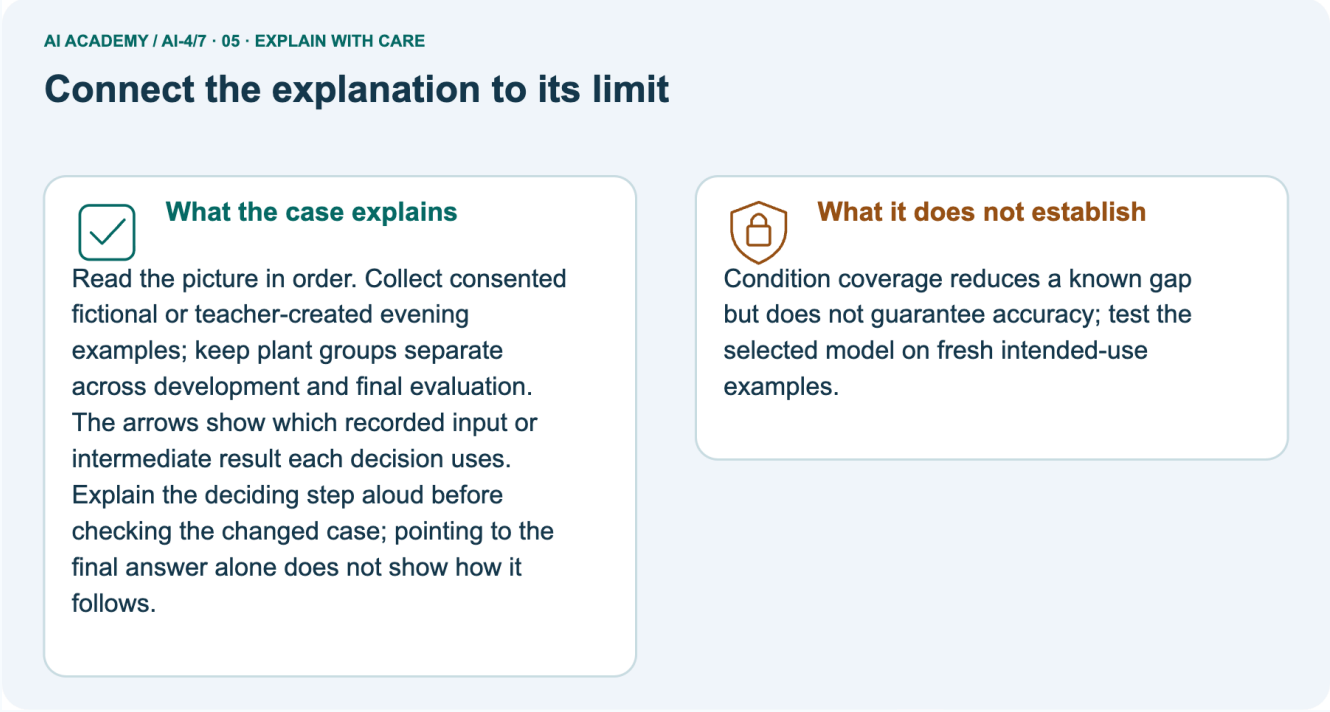
Original case: Collect consented fictional or teacher-created evening examples; keep plant groups separate across development and final evaluation.

Changed condition: Collect 12 more midday windowsill photos of the same plants. Expected result: The evening-coverage gap remains. Reason: More rows in the same condition do not supply **evidence** about the missing condition.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Collect consented fictional or teacher-created evening examples; keep plant groups separate across development and final evaluation.

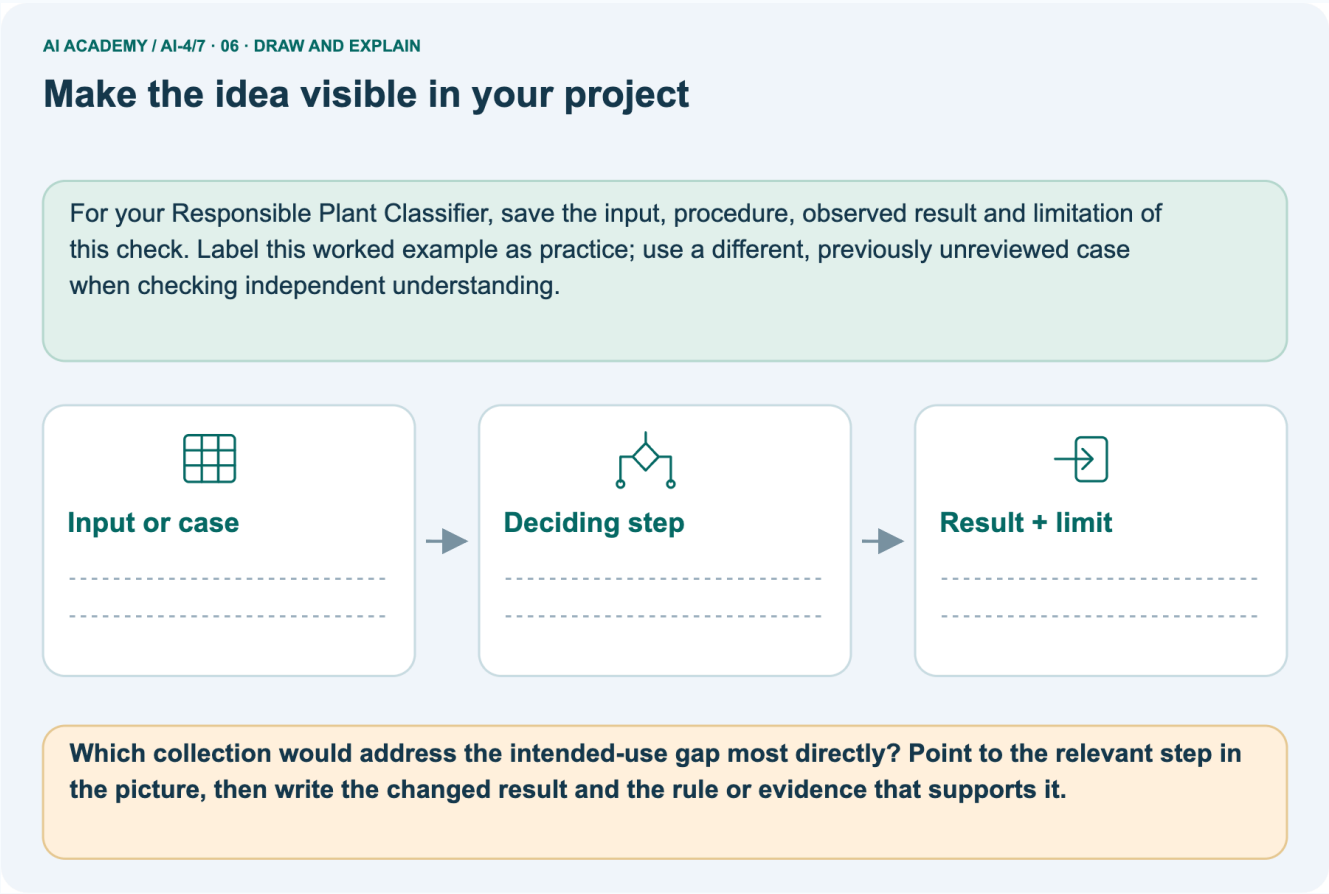
The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Condition coverage reduces a known gap but does not guarantee accuracy; test the selected model on fresh intended-use examples.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which collection would address the intended-use gap most directly? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.